

Notes on lecture 19

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1 Stokes theorem continued ...

Remark 1.0.1. Whenever you are given a problem in which you have to use Stokes theorem, you should look at the following: From the inputs provided to you, calculating the line integral is going to be simpler or the surface integral. Depending upon whichever is simpler, you should proceed. Sometimes, it may so happen that even without using Stokes theorem, you can evaluate the integral in a much straight forward way.

Example 1.0.2. Evaluate $\int_S \int (\nabla \times \vec{F}) \cdot d\vec{S}$ where $\vec{F} = xz\hat{i} + yz\hat{j} + xy\hat{k}$, and S is the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane. Figure 1.0.1 shows the surface S in blue color.

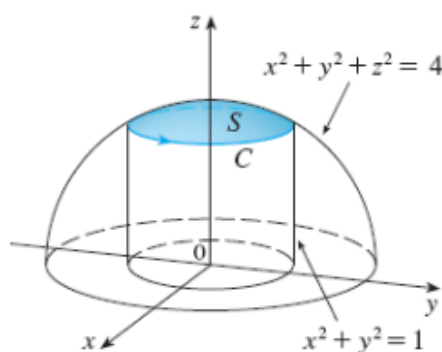


Figure 1.0.1: Figure for Example 1.0.2

By inspection, we can see that evaluating the surface integral is going to be

cumbersome. It is perhaps better to convert the integral into a line integral. By Stokes theorem, we have

$$\int_S \int (\nabla \times \vec{F}) \cdot d\vec{S} = \oint_C \vec{F} \cdot d\vec{r},$$

where C is a positively oriented curve that bounds the surface S . We have to find out such a curve C here. Consider the curve of intersection of the sphere with the cylinder. This curve certainly bounds the surface S . If we substitute $x^2 + y^2 = 1$ in the equation $x^2 + y^2 + z^2 = 4$ of the sphere, then we will get $z^2 = 3$, which in turn implies that $z = \sqrt{3}$ [We have taken the positive square root of 3 because the surface is above the xy -plane.]. If we choose this curve of intersection to be C , and take the anti-clockwise orientation on it, then we get a curve which is positively oriented and bounds the surface S .

The equations for the curve C are given by:

$$x^2 + y^2 = 1, \quad z = \sqrt{3}.$$

A parametrization of C is given by:

$$\vec{r}(t) = \langle \cos t, \sin t, \sqrt{3} \rangle, \quad 0 \leq t \leq 2\pi.$$

So by Stokes theorem, we have

$$\begin{aligned} \int_S \int (\nabla \times \vec{F}) \cdot d\vec{S} &= \oint_C \vec{F} \cdot d\vec{r} = \int_0^{2\pi} \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt \\ &= \int_0^{2\pi} \langle \sqrt{3} \cos t, \sqrt{3} \sin t, \cos t \sin t \rangle \cdot \langle -\sin t, \cos t, 0 \rangle dt \\ &= \int_0^{2\pi} (-\sqrt{3} \sin t \cos t + \sqrt{3} \sin t \cos t) dt = 0. \end{aligned}$$

□

Remark 1.0.3. Please refer to the last remark of the last lecture. It has some significance. We are going to consider some problems related to it in the tutorial sheet. Whenever we consider a surface, in order to find a curve which bounds the surface, what are we going to take/choose? This choice plays a crucial role in some problems.

In the above example, even without doing all those calculations, we could have reached the same conclusion (that the value of the integral is 0), by choosing the curve C bounding the surface S in a specific way. You will find a couple of problems in the tutorial sheet related to this matter.

2 The divergence theorem

In this section, we are going to study the Divergence theorem, which is due to Gauss, and is also sometimes called the Gauss's divergence theorem.

At this point, recall the vector form of Green's theorem. Green's theorem was considered on a planar region (2-dimensional) and we had found that

$$\oint_C \vec{F} \cdot \hat{n} ds = \int_D \int (\nabla \cdot \vec{F}) dA.$$

This was the result in the vector form of Green's theorem. All this was in a 2-dimensional plane.

It makes sense to extend this result (Green's theorem) to the 3-dimensional space. Then, intuitively speaking, the line integral $\oint_C \vec{F} \cdot \hat{n} ds$ should have become a surface integral of the form $\int_S \int (\vec{F} \cdot \hat{n}) dS$ (for some surface S), and the double integral $\int_D \int (\nabla \cdot \vec{F}) dA$ should have become a triple integral of the form $\int \int_E \int (\nabla \cdot \vec{F}) dV$ for some solid region E in $\mathbb{R}^3$. It so happens that our intuition becomes true. The equation

$$\int_S \int (\vec{F} \cdot \hat{n}) dS = \int \int_E \int (\nabla \cdot \vec{F}) dV$$

is precisely the result stated in the Divergence theorem.

Theorem 2.0.1. Divergence theorem: *Let E be a simple solid region and let S be the boundary surface of E , given with positive (outward) orientation. Let $\vec{F}$ be a vector field whose component functions have continuous partial derivatives on an open region that contains E . Then*

$$\int_S \int \vec{F} \cdot d\vec{S} = \int \int_E \int \operatorname{div} \vec{F} dV.$$

Remark 2.0.2. In the statement of Divergence theorem, E is a simple solid region. That is, E can be simultaneously considered as a type 1 or 2 or 3 region in $\mathbb{R}^3$. Even if E is not a simple solid region, we can definitely express E as the union of some simple solid regions. Then by breaking up the integral, we can prove the divergence theorem.

PROOF: (of Divergence theorem for a simple solid region:) Assume that $\vec{F} = \langle P, Q, R \rangle$ where P, Q, R are functions of x, y, z having continuous

partial derivatives. Then the left hand side (LHS) integral in the statement of divergence theorem equals

$$\int_S \int (\vec{F} \cdot \hat{n}) dS = \int_S \int (P\hat{i}) \cdot \hat{n} dS + \int_S \int (Q\hat{j}) \cdot \hat{n} dS + \int_S \int (R\hat{k}) \cdot \hat{n} dS$$

and the right hand side (RHS) integral in the statement of divergence theorem equals

$$\int \int_E \int \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dV.$$

If we can prove that

$$\int_S \int (P\hat{i}) \cdot \hat{n} dS = \int \int_E \int \frac{\partial P}{\partial x} dV, \quad (1)$$

$$\int_S \int (Q\hat{j}) \cdot \hat{n} dS = \int \int_E \int \frac{\partial Q}{\partial y} dV, \quad (2)$$

and

$$\int_S \int (R\hat{k}) \cdot \hat{n} dS = \int \int_E \int \frac{\partial R}{\partial z} dV, \quad (3)$$

then we are done.

We will prove (3). The proofs of (1) and (2) are similar. To prove (3), we are going to consider E as a type 1 region. That is,

$$E = \{(x, y, z) \in \mathbb{R}^3 | (x, y) \in D \subseteq \mathbb{R}^2, u_1(x, y) \leq z \leq u_2(x, y)\}.$$

Figure 2.0.1 shows E as a type 1 region.

Now the RHS integral in equation (3) equals

$$\begin{aligned} \int \int_E \int \frac{\partial R}{\partial z} dV &= \int_D \int \left[\int_{z=u_1(x,y)}^{u_2(x,y)} \frac{\partial R}{\partial z} dz \right] dA \\ &= \int_D \int [R(x, y, u_2(x, y)) - R(x, y, u_1(x, y))] dA. \end{aligned} \quad (4)$$

And the LHS integral in equation (3) equals $\int_S \int (R\hat{k}) \cdot \hat{n} dS$.

The region E is bounded by the surfaces S_1 , S_2 and S_3 (as shown in Figure

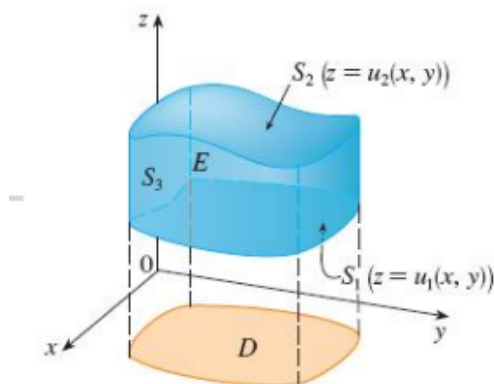


Figure 2.0.1: E as a type 1 region

2.0.1). The surface S_1 is the lower one, S_2 is the upper one, and S_3 is the side surface. The surface S is the union of S_1 , S_2 and S_3 . So we have

$$\int_S \int (R\hat{k}) \cdot \hat{n} dS = \sum_{i=1}^3 \int_{S_i} \int (R\hat{k}) \cdot \hat{n} dS.$$

Let us focus on $\int_{S_3} \int (R\hat{k}) \cdot \hat{n} dS$ first. On S_3 , observe that any outward normal $\hat{n}$ is always going to be horizontal. And the vector $\hat{k}$ is in the vertical direction. The dot product of $\hat{k}$ and $\hat{n}$ is going to be 0. Therefore we have

$$\int_{S_3} \int (R\hat{k}) \cdot \hat{n} dS = 0.$$

Let us now focus on $\int_{S_2} \int (R\hat{k}) \cdot \hat{n} dS$. Since S_2 is the upper surface, it follows that the normal $\hat{n}$ to it will have its z -component positive (that is, the outward normal will point upwards). In fact, the normal vector will be $\langle -\frac{\partial u_2}{\partial x}, -\frac{\partial u_2}{\partial y}, 1 \rangle$. We can then apply Remark 2.3.5 of the previous lecture notes, and conclude that

$$\int_{S_2} \int (R\hat{k}) \cdot \hat{n} dS = \int_D \int R(x, y, u_2(x, y)) dA.$$

Similarly, for the lower surface S_1 , the normal $\hat{n}$ will have its z -component negative (that is, the outward normal will point downwards). In fact, the

normal vector will be $\langle \frac{\partial u_1}{\partial x}, \frac{\partial u_1}{\partial y}, -1 \rangle$. So we can do a calculation similar to that shown in Remark 2.3.5 of the previous lecture notes, and conclude that

$$\int_{S_1} \int (R\hat{k}) \cdot \hat{n} dS = - \int_D \int R(x, y, u_1(x, y)) dA.$$

Therefore,

$$\int_S \int (R\hat{k}) \cdot \hat{n} dS = \int_D \int [R(x, y, u_2(x, y)) - R(x, y, u_1(x, y))] dA,$$

which is nothing but the expression on the right hand side of equation (4) above.

We can work out similarly for proving (1) and (2). For those proofs, we will have to consider E as a type 3 or type 2 region and proceed. This completes the proof. $\square$

Example 2.0.3. Find the flux of the vector field $\vec{F}(x, y, z) = z\hat{i} + y\hat{j} + x\hat{k}$ over the unit sphere $x^2 + y^2 + z^2 = 1$.

Basically, we have to find $\int_S \int \vec{F} \cdot d\vec{S}$, where S is the unit sphere. By divergence theorem,

$$\int_S \int \vec{F} \cdot d\vec{S} = \int \int \int \nabla \cdot \vec{F} dV,$$

where E is the solid unit sphere. Here,

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(z) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(x) = 1.$$

This implies that

$$\int \int \int \nabla \cdot \vec{F} dV = \int \int \int 1 dV = \text{Volume of } E = \frac{4}{3}\pi(1)^3 = \frac{4\pi}{3}.$$

So we could compute a complicated surface integral in a much simple way using divergence theorem. $\square$

Example 2.0.4. Evaluate $\int_S \int \vec{F} \cdot d\vec{S}$, where

$$\vec{F}(x, y, z) = xy\hat{i} + (y^2 + e^{xz})\hat{j} + \sin(xy)\hat{k}$$

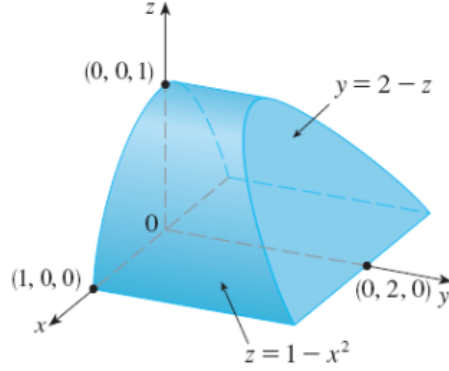


Figure 2.0.2: Figure for Example 2.0.4

and S is the surface of the region E bounded by the parabolic cylinder $z = 1 - x^2$ and the planes $z = 0$, $y = 0$, and $y + z = 2$. Figure 2.0.2 shows the region E in blue color.

Observe from the figure that if we take the projection/shadow of E on the xz -plane, then the computation of the integral will become easy. That means, we have to consider E as a type 3 region as follows:

$$E = \{(x, y, z) \in \mathbb{R}^3 \mid -1 \leq x \leq 1, 0 \leq z \leq 1 - x^2, 0 \leq y \leq 2 - z\}.$$

Observe also that

$$\begin{aligned} \nabla \cdot \vec{F} &= \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(y^2 + e^{xz^2}) + \frac{\partial}{\partial z}(\sin(xy)) \\ &= y + 2y + 0 = 3y. \end{aligned}$$

Then by divergence theorem,

$$\begin{aligned} \int_S \int \vec{F} \cdot d\vec{S} &= \int \int \int_E \nabla \cdot \vec{F} dV \\ &= \int_{x=-1}^1 \int_{z=0}^{1-x^2} \int_{y=0}^{2-z} 3y dy dz dx. \end{aligned}$$

The rest of the calculation is left to the reader as an exercise. □

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